

# Reproduction of *Compositional Visual Generation with Composable Diffusion Models*

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## ABSTRACT

We reproduce Liu et al.’s composable diffusion framework by adapting its zero-shot compositional inference to (1) a pretrained face generator (Stable Diffusion v1.5 + ControlNet) and (2) a controlled dataset. In both settings, we generate images conditioned on multiple attributes and measure correctness (via classifiers or IOU) and FID. Despite qualitative promise, composable inference fails to outperform vanilla sampling under limited compute, highlighting under-fitting and sparse text embeddings as key bottlenecks.

## 1 INTRODUCTION

Large text-guided diffusion models have demonstrated remarkable capability in generating photorealistic images from natural language descriptions [8–10]. However, these models frequently struggle with compositional understanding, often confusing attributes of different objects or their spatial relationships within complex scenes [5, 7]. This limitation stems from their reliance on fixed-size latent representations, which can hinder their ability to factorize and combine distinct conceptual elements [12].

To address this, [5] proposes an alternative structured approach. It reinterprets diffusion models as energy-based models (EBMs), allowing for the explicit combination of diffusion models via composable inference algorithm.

A key strength of this approach is its ability to generate significantly more complex scenes at inference. This method outperforms existing baselines, achieving, for instance, a 24.02% higher accuracy for three objects on the CLEVR dataset.

Despite its advancements, the method has limitations. It inherits its conceptual misunderstandings from its pre-trained diffusion models (e.g., misinterpreting “person” in GLIDE [7]) and can still confuse object attributes (e.g., “red bear” versus a “bear in a red forest”). Composition may also fail, particularly for objects positioned centrally or when dealing with highly abstract combinations [9].

### Initial thoughts about the paper

As a group we liked how general the method was, as it opened many avenues for the reproduction. We also like the general applicability and ease of the method. I thought the zero shot learning aspect of the method made it particularly elegant. We also thought the demonstrations of the method in the paper were very interesting, although computationally infeasible for us, but the wide applicability of the method made that easy to adapt.

## 2 BACKGROUND

The proposed composable inference algorithm combines Conditional Diffusion Models (CDM) and Energy-Based Models (EBMs) [1, 3]. CDMs operate by iteratively denoising a sample from pure Gaussian noise back to a sharp image [11, 12]. This process involves a forward stage that progressively adds Gaussian noise to data and

a reverse stage that learns to reverse this noisy process, typically by predicting the added noise or the clean image. The iterative nature of the reverse process allows for the generation of complex data distributions.

Energy-Based Models (EBMs), conversely, define a probability distribution over data using an unnormalized density function based on an energy function,  $p(x) = \frac{\exp(-E(x))}{Z}$  [4]. Sampling from an EBM is commonly achieved through iterative methods like Langevin dynamics, by following the gradient of the energy function [6].

A crucial insight of this work is the functional equivalence between the sampling procedures of CDMs and EBMs. The score function of the data distribution is directly proportional to the negative gradient of an energy function in EBMs [1]. This strong theoretical connection allows a trained diffusion model to be viewed as an implicitly parameterized EBM, allowing EBM-specific composition techniques to work with diffusion models [5].

The core of the composable inference involves generating images conditioned on a set of concepts  $\{c_1, c_2, \dots, c_n\}$ , where each concept is separately processed by the same diffusion model. This is achieved through compositional operators. The two operators are:

**Conjunction:** To combine multiple concepts, the paper leverages the property that the joint probability of independent events can be factorized. Expressed as a weighted sum of the individual concept scores and the unconditional score [5], the score is defined as  $\hat{\epsilon}(x_t, t) = \epsilon_\theta(x_t, t) + \sum_{i=1}^n w_i (\epsilon_\theta(x_t, t|c_i) - \epsilon_\theta(x_t, t))$ , where  $w_i$  are user-defined weights controlling the influence of each concept.

**Negation:** This operator allows for generating images that exclude a specific factor. It modifies the score function to bias sampling away from the undesired concept: formulated as  $\hat{\epsilon}(x_t, t) = \epsilon_\theta(x_t, t) + w(\epsilon_\theta(x_t, t|c_i) - \epsilon_\theta(x_t, t|\tilde{c}_j))$ .

The main claim of [5] is that applying compositional operators to diffusion models allows for instant (zero-shot) improvement in inference quality. This is measured using accuracy and FID score [2]. The same claim will be tested in this paper. The expected result is that as the number concepts in the prompts increase, the FID and Accuracy will decrease, but always be higher than the accuracy and FID of the same model with a vanilla inference algorithm.

## 3 REPRODUCTION

### 3.1 Approach

Large diffusion models are difficult to train. [5] trained all their models for multiple days on large GPU clusters, with the CLEVR or FFHQ Datasets (which are very large datasets) to report the results in their paper. Due to computational constraints, none of the reported results in the paper could be reproduced. Therefore, an alternate approach was followed, with the goal of verifying the claims made by the authors.

The main claim of [5] is that diffusion models can be composed by formulating them as implicitly parametrized energy based models. For our reproduction, this claim will be tested. The simpler approach is to test it with a pretrained diffusion model, with a modified inference algorithm (Section 3.1.1). The second way is to test it with a controlled dataset (Section 3.1.2). Both avenues of reproduction are explored in this paper.

**3.1.1 Base Reproduction.** The models used in [5] are unavailable to the public. However, one of the experiments conducted in the paper generated composable features on faces. Diffusion models trained on FFHQ are used to generate images of people with characteristics, such as glasses, gender, and hair color, which can then be checked with binary classifiers (trained on different facial datasets). This was the reproduction task of a fellow group member, Alexandru, and his code was used in this reproduction as well.

Diffusion models trained on facial data are rather common, so I will be using a pretrained facial diffusion model to generate faces, which I will then assess for correctness. This approach is similar to the last approach presented in [5].

**3.1.2 Controlled Reproduction.** Most of the models in [5] are trained on the CLEVR dataset. One of the main complexities of the CLEVR dataset is the variety of the objects, the fact that all the objects are placed in a 3D scene, and the size of the dataset. These all contribute to the high computational cost of training and testing a diffusion model. By simplifying these aspects of a dataset, a much simpler diffusion model can be trained, functioning as a "proof of concept" of the main claim of the paper, while being significantly more computationally feasible.

The controlled dataset has 5 shapes (Circle, Square, Triangle, Star, Hexagon), and 5 colors (Red, Green, Blue, Yellow, Black). In one image, up to 5 shapes (with one of the random colors) can be placed randomly. The label for one sample is the shape, its color, and its normalized position in the image. This information is used to construct the prompt (e.g. for Figure 1, the prompt would be Blue star at (0.4, 0.1), Blue Hexagon at (0.75, 0.35))

## 3.2 Method

**3.2.1 Base Reproduction.** The inference algorithm from [5] was adapted to 5 for use with a pretrained large scale diffusion model, Stable Diffusion v1.5 with ControlNet (SDC). Since Stable Diffusion is trained on a large variety of data, it should be able to generate images of faces as needed.

All possible combinations of 3 variables (Gender, Hair color, and Glasses) are iterated and 1000 images per prompt are generated, using different seed values. Both inference algorithms are used to generate images.

Then, for each of the images, a binary classifier (developed by Alexandru) is used to determine whether or not the output is sincere to the prompt. This served as a correctness measure for the generated images, and is reported as the accuracy measure.

**3.2.2 Controlled Reproduction.** A dataset consisting of 10,000 images, with 1,000 test images and their respective prompts, is generated, which serve as ground truth test cases to compare with.

Based on the dataset, prompts were constructed and embedded into latent vectors (one hot encoding for the shapes and colors, and

the coordinates were mapped to  $[0, 1]$  and put into the embedding vector). Then, a small diffusion model was trained for conditional image generation.

To test the model, the prompt from the test cases were embedded, then images were generated using the diffusion model (using a vanilla and composed inference algorithm), and compared to the original test cases using Intersection Over Union (IOU) as accuracy. This is reported as the accuracy value. Deviations from the distribution of training/test images and the generated images are also measured using FID.

## 3.3 Implementation details

SDC was loaded from HuggingFace, and run on Google Colab and Kaggle (based on available free GPUs), using a python notebook. 100 time steps were used to generate each image. It was run with the pipeline’s default noise scheduler, and a classifier free guidance scale of 7.5. Images are generated at  $512 \times 512$  resolution and scaled by the scheduler’s `init_noise_sigma`. The entire procedure was executed in half-precision.

A small diffusion model from the SmallDiffusion python library was trained with the controlled dataset for 300 epochs, 64 images per batch, with a resolution of  $64 \times 64$ . This configuration defines a 3-level UNet with a base channel size of 64, using channel multipliers (1, 1, 2) across levels, and applying attention at resolution 14. Conditional embeddings were incorporated to condition generation on the input prompt features. The training employed the Adam optimizer with a learning rate of  $1e-4$ , and mean squared error (MSE) was used as the reconstruction loss. To ensure consistency, a fixed noise schedule was adopted across all runs. In total, training the model took 14 hours on a Nvidia 1050Ti with 4GB VRAM. After training, the model’s generative quality was assessed both qualitatively (via visual inspection of sampled outputs) and quantitatively, using Fréchet Inception Distance (FID) scores computed against the test set. For ease of training, a template diffusion model (originally trained on FashionMNIST) was trained from scratch.

In coding the composable diffusion algorithm, the codebase for [5] was used as a reference, since it could not be used directly due to computational constraints.

## 4 RESULTS AND DISCUSSION

### 4.1 Base Reproduction

#### Qualitative

Despite being as close to the algorithm in [5] as possible, the generated results have a large difference in quality (Figure 2). Without FID scores it is difficult to comment exhaustively on the quality of the results, but it appears as if all the prompted features are present in the image. Furthermore, some experimentation with concept weights revealed that the magnitude of the weights heavily impact the quality of the generated images. For the final inference, the weight of the concepts was maintained at 1.0 for consistency.

#### Quantitative

The accuracy of the models are not high, even for 1 component (Table 1). Since the model is not explicitly trained on facial datasets, it makes sense. As the number of components increase, the accuracy drops, indicating that the prompt embedding is not

| Models         | 1 Component<br>Acc (%) ↑ | 2 Components<br>Acc (%) ↑ | 3 Components<br>Acc (%) ↑ |
|----------------|--------------------------|---------------------------|---------------------------|
| Vanilla SDC    | 68.12                    | 48.84                     | 23.13                     |
| Composable SDC | 45.20                    | 9.14                      | 1.05                      |

**Table 1:** Accuracy evaluation of SDC using vanilla and composable inference

sufficient to generate composed images (this was also the main problem introduced in [5]). However, the composable inference does not improve the accuracy. As expected, the accuracy of the models drop as the number of components increase, but the drop is very large. With 3 components, the accuracy drops to 1%. Therefore composable inference does not beat the vanilla inference, and does not enable successful composition of facial features. The success of either model with 1 component is likely due to the effectiveness of CLIP at creating textual embeddings. This decline suggests the prompt embeddings generated by CLIP may be insufficiently expressive for multi-attribute composition, particularly in domains (like facial attributes) not seen jointly during training.

## 4.2 Controlled Reproduction

### Qualitative

In Figure 4 we can see the output of the diffusion model for unseen test cases. Even after training for 14 hours, it is clear that there needs to be more training for the model to obtain sharper results. There also does not seem to be a variation in the number of shapes across the images, and the colors are also mixed in some of the shapes.

### Quantitative

Even though the results from [5] are not fairly comparable to the results in Table 2, there is a stark difference in the values. The range of FID values in Table 1 of [5] are less than 80 (for 1 component), while the FID values in Table 2 are greater than 100. The accuracy rates are also low, and its effects can be seen in Figure 4, where the number of shapes seems to be consistent over the images, even though the prompts have a random number of shapes. Unlike the results reported in [5], the FID remains constant as the number of components increase. In conjunction with the accuracy dropping steeply, this points to a critical issue with training the diffusion model. The high FID and low accuracy suggest the model underfits the training data. As a result, the model is not able to generate images with vanilla or composable inference. Surprisingly, the FID is higher and the accuracy lower with composable inference. This is likely due to the noise estimates being of lower quality to begin with, which degrades further in quality when composed. This aligns with the theoretical expectation that composable inference relies on accurate conditional gradients. Since the base model already produces blurry outputs, the composed result is even less reliable. The decreasing accuracy also points to the model not understanding the given prompt. Therefore, the issue is most likely with the model not understanding the prompt, which would be the case when using a sparse encoding for the prompts.

## 5 CONCLUSION

Our experiments clarify which adjustments—principally tighter hyper-parameter logging and lighter-weight schedulers—could bring a follow-up attempt within reach of a single GPU. That being said, reproducing the claims of the paper also proved to be a challenge. Of the many factors affecting the quality of the reproduction, the main were the lack of computational resources and the difficulty in adapting the inference algorithm to run on free and low-end computational resources. Normally, writing a new inference algorithm from scratch (as we initially intended) would require writing a lot of code and computational resources to properly test and debug it. Not having worked with diffusion models at any level before, we did not realize this issue with our approach. This meant that existing code (from Hugging Face) had to be adapted, which introduced other difficulties, since SDC uses a lot of optimization techniques in the backend to speed up image generation and increase sampling quality. Therefore, when writing custom inference algorithms, the quality instantly drops, since we are unable to make use of the same optimizations as the backend. Lastly, the lack of computational resources also affected our ability to evaluate FID scores for SDC, as doing so would require large datasets, which are too large to compute, especially taking into account time constraints. A rough yet informative proxy e.g., CLIP-score or FID with an Inception-FD15 backbone—could have offered quantitative insight without exceeding memory limits.

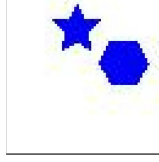
Even with two approaches, the reproduction of the claims in [5] were not successful. This was for a variety of reasons, but mostly had to do with computational resources and the used libraries. It is evident from the results of the controlled experiment that the model was unable to learn conditional image generation (due to sparse text embeddings). Exploring dense conditioning schemes such as FiLM-style layers or T5 embeddings may resolve this bottleneck without increasing model size. This is something that can be explored using more dense text embedding methods, and might serve as a more accessible avenue for reproduction. Due to computational and time constraints, the model could not be trained again. Longer training on the same GPU could narrow the under-fitting gap while respecting identical hardware limits.

## REFERENCES

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| Models               | 1 Component |        | 2 Components |        | 3 Components |        |
|----------------------|-------------|--------|--------------|--------|--------------|--------|
|                      | Acc (%) ↑   | FID ↓  | Acc (%) ↑    | FID ↓  | Acc (%) ↑    | FID ↓  |
| Vanilla Diffusion    | 52.14       | 101.45 | 14.54        | 98.54  | 3.85         | 104.32 |
| Composable Diffusion | 45.20       | 115.94 | 9.14         | 118.84 | 1.05         | 116.48 |

**Table 2:** Evaluation of Diffusion Model using vanilla and composable inference algorithm



**Figure 1:** Sample from the controlled dataset

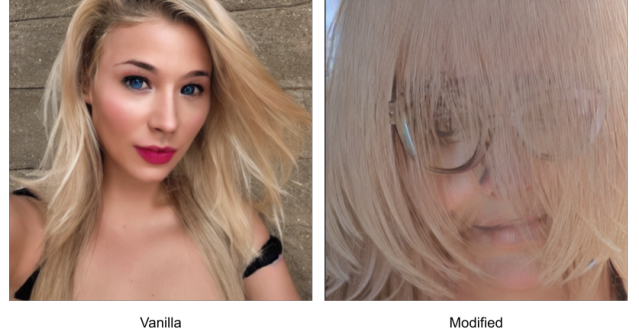
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## A INDIVIDUAL CONTRIBUTION STATEMENT

I adapted the paper’s compositional-inference code to a public pipeline (SDC), generated 1 000 images for every combination of gender, hair-color and glasses with 100 denoising steps, and benchmarked them with an external facial-attribute classifier to quantify accuracy drops under composition. I also designed a controlled reproduction: synthesising a 10 000 image dataset of colored geometric shapes, then training a 64×64 conditional diffusion model and evaluating it with FID and accuracy to diagnose underfitting and weight sensitivity. Everything developed in this paper was done by me, except the binary Facial Attribute classifiers.

## B SAMPLE OF SHAPE DATASET

## C DEPICTION OF NORMAL AND CUSTOM INFERENCE STEP



**Figure 2:** Vanilla inference vs. Composable (Modified) inference

## D IMPACT OF CUSTOM WEIGHTS ON GENERATED IMAGES



**Figure 3:** Difference in image quality as the magnitude of weights increase, the used prompt was *A woman with glasses and hair*. The numbers in the image denote the weight per concept of the prompt

## E TEST OUTPUT OF DIFFUSION MODEL



**Figure 4:** Test prompts generated using the controlled dataset diffusion model

## F MODIFIED INFERENCE ALGORITHM

```
for t in pipe.scheduler.timesteps:
    # (3.1) duplicate the latents so UNet sees every conditioning batch
    latent_in = latents.repeat(n_total, 1, 1, 1)
    latent_in = pipe.scheduler.scale_model_input(latent_in, t)

    # (3.2) UNet forward for *all* prompts in one go
    noise_pred = pipe.unet(latent_in, t,
                           encoder_hidden_states=text_embeds).sample

    # (3.3) guidance-by-composition
    noise_pred_uncond = noise_pred[0]          #  $\epsilon$ 
    j, eps = 1, noise_pred_uncond.clone()

    # conjunction terms           $\epsilon \ += \ w \ (\epsilon_i - \epsilon)$ 
    for w in w_pos:
        eps += w * (noise_pred[j] - noise_pred_uncond)
        j += 1

    # negation pairs           $\epsilon \ += \ w \ (\epsilon_{pos} - \epsilon_{neg})$ 
    for w in w_neg:
        eps += w * (noise_pred[j] - noise_pred[j + 1])
        j += 2

    # (3.4) scheduler step
    latents = pipe.scheduler.step(eps, t, latents).prev_sample

imgs = pipe.vae.decode(latents / pipe.vae.config.scaling_factor).sample
img = (imgs[0] / 2 + 0.5).clamp(0, 1).cpu().permute(1, 2, 0).numpy()
return Image.fromarray((img * 255).astype("uint8"))
```

**Figure 5:** Relevant portion of the modified inference code, as used in the Base and Controlled reproduction

## G AI STATEMENT

I certify that no artificial intelligence tools or systems were used in the creation, editing, or revision of this report; all work (code and report) is entirely my own original effort.

## H CODE REPOSITORY

All code can be found on [GitHub](#)